

Use of Mental Health Care and Substance Abuse Treatment Among Adults With Co-occurring Disorders

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Objectives: This study investigated patterns of use of mental health care and substance abuse treatment for a nationally representative sample of adults with co-occurring mental health problems and a substance use disorder and compared these patterns with those of persons with either a mental health problem or a substance use disorder. **Methods:** Data were from the 2001 and 2002 National Surveys on Drug Use and Health. The study examined rates of substance use disorders and mental health problems among adults aged 18 years and older, rates of substance use disorders among adults with mental health problems, and rates of mental health problems among adults with substance use disorders. Next, rates of substance abuse treatment and mental health care use were calculated among five groups that were formed on the basis of the presence of a substance use disorder, mental health problems, or both in the past year. **Results:** A total of 2,851 respondents had a substance use disorder only, 1,633 had a substance use disorder with one or more mental health symptoms and without serious mental illness, 1,872 had a substance use disorder with serious mental illness, 13,759 had one or more mental health symptoms only, and 7,530 had a serious mental illness only. A substantial proportion of adults with comorbid mental health problems and a substance use disorder did not receive any treatment (46 percent of those with serious mental illness and 65 percent of those with one or more mental health symptoms). Co-occurring substance use disorder was not associated with increased use of mental health care. The likelihood of receiving any substance abuse treatment increased with the presence and severity of mental health problems. Across all five groups, use of mental health care was more common than use of substance abuse treatment. Less than one-third of patients with comorbid mental health problems and a substance use disorder who used mental health care also received substance abuse treatment. **Conclusions:** The large proportion of untreated individuals with mental and substance use disorders reinforces existing concerns about barriers to beneficial treatment. Low rates of use of substance abuse treatment among patients who have comorbid mental health problems and a substance use disorder and use mental health care suggest that recommendations that substance use disorders be treated before, or concurrently with, mental disorders have not been widely adopted. (*Psychiatric Services* 56:954–959, 2005)

The high rate at which mental and substance use disorders occur together has been well documented in epidemiologic and clinical studies (1–4). In the National Comorbidity Study (NCS), roughly half of respondents who met criteria for a substance use disorder at some time in their life also met criteria for one or more lifetime mental disorders (2). Similarly, half of those who met criteria for a mental disorder at some time in their life also met criteria for a lifetime substance use disorder (2).

Despite these high rates, relatively little is known about the patterns of use of substance abuse treatment and mental health care among the population with co-occurring substance and mental health problems. In general, sample size restrictions in epidemiologic surveys have limited researchers' ability to examine the joint and conditional probabilities of service use by the presence and severity of co-occurring disorders. At the same time, a lack of diagnostic information limits the utility of administrative claims data for this purpose.

Nonetheless, studies that use population-based surveys to examine adults with co-occurring mental and substance use disorders reveal a number of patterns. First, between 70 and 80 percent of individuals with co-occurring mental and substance use disorders did not receive any mental health care or substance abuse treatment in the past year (5). Second, only 8 percent reported using integrated care or even both types of care sometime in the same year (5). Third, among those who used a single serv-

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ice, use of mental health care was more than three times as common as use of substance abuse treatment (5). Finally, co-occurring mental and substance use disorders were associated with an increased likelihood that someone with a substance use disorder would receive substance abuse treatment or participate in a self-help group (6,7).

National surveys of providers of substance abuse treatment show that roughly half of all centers offer special programs for clients with dual diagnoses and that this proportion has remained fairly stable over the past decade (8,9). At the same time, there is evidence that the rate at which privately owned treatment centers refer patients with severe mental illness to other health care providers has increased over this period (8). Unfortunately, it is not known whether or not these referrals are to providers who offer integrated treatment programs or mental health care only.

In this article we provide new and updated descriptive information on patterns of use of mental health care and substance abuse treatment for a nationally representative sample of adults with co-occurring mental health problems and substance use disorders. We compared these service use patterns with those of adults with substance dependence only or mental health problems only. Our analysis used data from the 2001 and 2002 National Survey on Drug Use and Health (NSDUH, formerly the National Household Survey on Drug Abuse) (10,11). The NSDUH offers several important advantages in investigating this type of question. As with the NCS (12), the Health Care for Communities survey (HCC) (13), and the National Epidemiologic Survey on Alcohol and Related Conditions (NESARC) (14,15), the NSDUH used a complex, multistage probability sampling strategy and, therefore, can be used to calculate nationally representative estimates. However, the pooled data from the NSDUH contains a considerably larger sample of adults (N=90,277) compared with the NCS (N=8,098), the HCC (N=9,585), the NESARC (N=43,093) or the Epidemiologic Catchment Area study (N=20,861)

(16). This extremely large sample is especially advantageous for doing analyses of smaller subgroups, as is the case in the study presented here. The NSDUH employed items from structured clinical interviews, as described below, although the mental health diagnostic information is not as rich as it is in the ECA, the NCS, or the NESARC.

Methods

Data

Data came from the 2001 and 2002 NSDUH, a survey conducted annually by the Substance Abuse and Men-

pleted through an in-person interview that is computer assisted. Detailed information on data collection procedures and sampling design is available elsewhere (10,11).

Respondents were considered to have a substance use disorder if they reported three or more of the seven criteria for drug or alcohol dependence as specified in the American Psychiatric Association's *DSM-IV*. Our analysis focused on the subpopulation of dependent adults because of our interest in service use and the growing use of treatment guidelines and placement criteria that limit access to treatment for patients with relatively severe problems. For example, recently issued guidelines by the Department of Veterans Affairs and the Department of Defense recommend that individuals who screen positive in general medical settings for dependence be referred to specialty-based treatment (17) and those with less severe problems receive brief counseling.

In the absence of a standard for measuring objective mental health need, (18–20) we employed two distinct and complementary measures. First, the presence of mental health symptoms among respondents aged 18 years and older was measured by using summary symptom questions drawn from the Composite International Diagnostic Interview–Short Form (CIDI-SF), which was adapted to screen for symptoms of nonaffective psychosis (21–24). The mental health symptoms covered in the surveys represent key constructs from the major disorders (major depressive disorder, mania, generalized anxiety disorder, panic attacks, social phobia, agoraphobia, and posttraumatic stress disorder). Second, we used a clinically validated indicator of serious mental illness in the past year (21). The screener, also known as the K6, was developed for use in the U.S. National Health Interview Survey and was subsequently included in the NSUDH. The measure was intended to screen for serious mental illness, which was defined in Public Law 102-321 as having at least one 12-month *DSM-IV* disorder (excluding substance use disorders) along with functional impairment. As a result, the measure is

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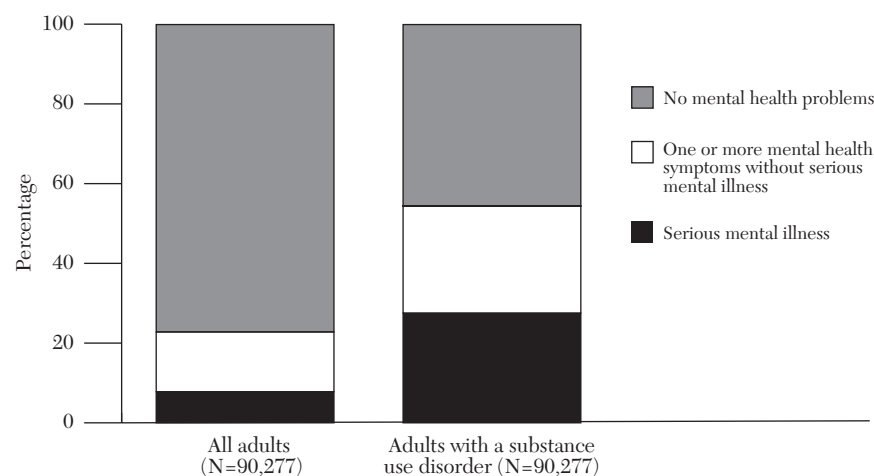
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tal Health Services Administration for the primary purpose of estimating the prevalence of illicit drug, alcohol, and tobacco use in the United States (10,11). The pooled sample included 137,055 respondents aged 12 years and older and had a combined response rate of 75 percent (10,11).

The survey takes roughly one hour to complete. To ensure confidentiality, questions about substance use, mental health problems, and treatment are completed through audio-assisted computer interview technology in which respondents key their responses directly into a laptop computer. Remaining questions are com-

Figure 1

Past-year mental health problems among adults surveyed by the 2001 and 2002 National Surveys on Drug Use and Health^a



^a Unweighted sample sizes

substantially more inclusive than other serious mental illness measures that rely on a previous psychiatric inpatient stay or a diagnosis of schizophrenia or other forms of psychosis.

The K6 is based on six questions that measure how frequently respondents experience symptoms of psychological distress (that is, feelings of nervousness, hopelessness, restlessness, depression, and worthlessness, and feeling as though everything is an effort) during the one month in the past year when they were feeling at their worst emotionally. Scale values range from 0 to 24, with a score of 13 or above indicating a high likelihood of serious mental illness. Both the modified CIDI-SF questions and the K6 items were structured such that positive responses were recorded if respondents indicated that specified symptoms were present any time during the past 12 months.

The survey asked a series of questions about the use of substance abuse treatment. Respondents were first asked whether they have received treatment or counseling for their use of alcohol or any drug, not including cigarettes, in the past 12 months, including detoxification and any other treatment for medical problems associated with drug or alcohol use. Next, respondents who answered affirmatively were asked about the

form of treatment. We excluded individuals who reported either participation in self-help groups or brief counseling in emergency departments as their only form of treatment to better ensure that our results inform issues of access to and use of the paid care system. We would note that our results were virtually unchanged when our definition of treatment was broadened to include self-help groups, because a very small percentage of individuals in the NSDUH were dependent on substances in the past year used self-help as their only form of care.

Respondents were considered to use mental health care if they reported during the past 12 months having an inpatient stay for mental health treatment in a general or psychiatric hospital; receiving care in a mental health clinic; receiving mental health treatment in an office of a physician, private therapist, psychologist, psychiatrist, social worker, or counselor; receiving mental health treatment in a partial day hospital, day treatment program, or other location, such as a church, shelter, or school; or taking any medication prescribed to treat a mental or emotional condition. Throughout the section of the interview involving mental health care use, care was defined as "treatment or counseling for problems with emo-

tions, nerves, or mental health," and respondents were explicitly asked to exclude treatment for alcohol and drug use when answering questions about use of mental health care.

Analyses

First, we calculated rates of substance use disorders and mental health problems among all adults aged 18 years and older, rates of substance use disorders among adults with mental health problems, and rates of mental health problems among adults with substance use disorders. Next we calculated rates of substance abuse treatment and mental health care use among five groups that were formed on the basis of the presence of a substance use disorder, mental health problem, or both in the past year: substance use disorder only (N=2,851), substance use disorder with one or more mental health symptoms and without serious mental illness (N=1,633), substance use disorder with serious mental illness (N=1,872), one or more mental health symptoms only (N=13,759), and serious mental illness only (N=7,530).

All proportions presented in descriptive tables were nationally weighted. Standard errors and difference in means tests were adjusted to account for the NSDUH's complex sampling design by using the survey analysis commands in STATA version 8.0 (25).

Results

Substance use and mental disorders

Substance use disorders were substantially more common among respondents with mental health problems than among respondents without such problems. Our study found that 4.3 percent of all respondents had a substance use disorder only, whereas the rate was 7.7 percent among those with one or more mental health symptoms and 15.1 percent among those with serious mental illness. At the same time, mental health problems were more common among adults with substance use disorders. Figure 1 shows that roughly half of adults with a substance use disorder have mental health problems, compared with roughly a quarter of all

Table 1

Past-year use of substance abuse treatment and mental health care among adults surveyed by the 2001 and 2002 National Surveys on Drug Use and Health, by type of problem^a

Service use	Substance use disorder						Mental health problems only			
	No mental health symptoms (N=2,851)		One or more mental health symptoms without serious mental illness (N=1,663)		Serious mental illness (N=1,872)		One or more mental health symptoms without serious mental illness (N=13,759)		Serious mental illness (N=7,530)	
	%	SE	%	SE	%	SE	%	SE	%	SE
Any treatment	20.7	1.5	35.0	2.0	54.1	2.0	22.9	.6	47.7	1.0
Substance abuse treatment only	7.8	.9	7.6	1.2	4.1	.6	.6	.1	.7	.1
Mental health care only	9.2	1.2	20.7	1.8	34.4	1.9	21.8	.6	45.5	1.0
Substance abuse treatment and mental health care	3.7	.7	6.6	1.0	15.5	1.7	.5	.1	1.4	.2
Any substance abuse treatment	11.4	1.1	14.2	1.5	19.3	1.9	1.1	.2	2.1	.3
Any mental health care	12.8	1.2	27.3	1.9	49.8	2.2	22.2	.6	46.8	1.0
Substance abuse treatment conditional on mental health care	28.5	5.0	24.2	3.4	31.2	3.0	2.4	.5	3.0	4.8
Mental health care conditional on substance abuse treatment	32	5.0	46.4	5.8	78.9	3.3	46.7	6.8	68.2	5.1

^a All proportions are nationally weighted.

adults. The prevalence of serious mental illness was more than three times higher among adults with substance use disorders than among all adults (27.5 percent compared with 7.8 percent).

Any service use

As shown in Table 1, substantial proportions of all diagnostic groups used no services of any kind in the past year. Among respondents with mental health problems only, 22.9 percent of those with one or more symptoms and 47.7 percent of those with serious mental illness used any services. Among respondents with substance use disorders, 20.7 percent with no mental health problems, 35.0 percent with one or more mental health symptoms, and 54.1 percent with serious mental illness used any services.

Substance abuse treatment

A minority in each of the three diagnostic groups with substance use disorder used substance abuse treatment either alone or in conjunction with mental health care. At the same time, the use of substance abuse treatment was strongly associated with co-occurring mental health problems. Among respondents with substance use disorders, only 11.4 percent of those with

no mental health problems used any substance abuse treatment, compared with 14.2 percent of those with one or more mental health symptoms and 19.3 percent of those with serious mental illness. These increasing rates were due to an increase in the proportion of people with mental health problems who received both mental health and substance abuse treatment rather than substance abuse treatment only. Compared with respondents with a substance use disorder and one or more mental health symptoms, respondents with a substance use disorder and serious mental illness were half as likely to use substance abuse treatment only (7.6 percent compared with 4.1 percent, $p < .05$) and more than twice as likely to use both substance abuse treatment and mental health care (6.6 percent compared with 15.5 percent, $p < .001$). Even among respondents with a substance use disorder and serious mental illness who received mental health treatment, only 31.2 percent received substance abuse treatment.

Mental health treatment

Among individuals with a substance use disorder only, use of mental health care was at least as common

as use of substance abuse treatment, with 12.8 percent receiving mental health care, compared with 11.4 percent receiving substance treatment. Interestingly, in the group with substance use disorder only the rate of mental health care use was comparable to the rate of 12.1 percent in the general population. The presence or absence of a substance use disorder was not strongly associated with the use of mental health care among those with mental health problems. Among those with one or more mental health symptoms, the use of mental health care was more common among those with a co-occurring substance use disorder than among those with mental health symptoms only (27.3 percent compared with 22.2 percent, $p < .05$). Almost half the respondents with serious mental illness received mental health care, regardless of the presence of substance use disorder (49.8 percent of those with substance use disorder compared with 46.8 percent of those without such a disorder). Mental health care was relatively frequent among respondents who received substance abuse treatment, with rates ranging from 32.0 to 78.9 percent for the various diagnostic categories.

Discussion

Several of our findings were consistent with those found in other population-based surveys, including the National Household Survey on Drug Abuse and the HCC, both from the late 1990s (5,6). First, we found that a majority of individuals in almost all diagnostic categories received neither mental health nor substance abuse treatment. The exception to this finding was among persons with a co-occurring substance use disorder and serious mental illness, of whom slightly more than half received either mental health or substance abuse treatment or both. Second, only a small number of individuals with comorbid mental and substance use disorders received treatment for both substance abuse and mental health problems in the past year. Possible reasons for low rates of treatment discussed in the literature include stigma, denial, financial barriers, failure to perceive need for services, a shortage of trained providers, faulty diagnoses, and the lack of a strong clinical consensus about the best way to treat patients with comorbid disorders (26,27).

Among individuals with substance use disorders, those with co-occurring mental and substance use problems were more likely to receive substance abuse treatment than those with a substance use disorder only. This pattern of treatment occurs because the rate of combined mental and substance abuse treatment is dramatically higher among such individuals (the rate at which these individuals receive only substance abuse treatment actually drops), suggesting that the mental health symptoms may cause individuals to seek mental health treatment, which results in substance abuse treatment in some cases. On the other hand, among individuals with mental health problems, those with co-occurring mental and substance use disorders were about as likely to receive mental health treatment as those with mental health problems only. Several characteristics of individuals with comorbid mental and substance use disorders may explain this lack of relationship: a relative lack of financial resources

with which to pay for mental health care, increased stigma associated with mental health care, or an expectation that mental health providers would require them to stop using substances.

We found that individuals in all diagnostic categories, even those with a substance use disorder only, were more likely to receive mental health treatment than substance abuse treatment. Low rates of use of substance abuse treatment relative to use of mental health care may reflect several factors. First, both public and private health insurance plans generally cover mental health care more generously than substance abuse treatment. Second, survey respondents may not understand the practical difference between the two types of care, despite the fact that the survey instructs respondents to distinguish among the reasons for seeking each type of care.

Particularly noteworthy were low rates of use of substance abuse treatment among adults with comorbid mental and substance use disorders who received mental health care. Only 31.2 percent of mental health care users with serious mental illness and a substance use disorder received any substance abuse treatment. This low rate of substance abuse treatment among persons who use mental health care raises policy concern, because mental health treatment alone for co-occurring disorders may be ineffective (28). Our results suggest that this recommendation has not been widely adopted by mental health care providers.

It is interesting to contrast the rates of use of substance abuse treatment among persons who receive mental health care with the rates of use of mental health care among those who receive substance abuse treatment. Relatively high rates of use of mental health care among persons with comorbid mental and substance use disorders who receive substance abuse treatment may reflect a greater tendency of providers of substance abuse treatment to refer patients with comorbid disorders for mental health care, which is reimbursed more generously by third-party payers and repre-

sents a primary focus of roughly one-third of substance abuse treatment providers (9).

Conclusions

Our findings highlight concerns about low overall treatment rates for individuals with a substance use disorder or mental problems or both and the relatively large role that the mental health care system plays in treating individuals with co-occurring mental health problems and a substance use disorder. Although rates of substance abuse treatment use increased with the receipt of mental health care, rates of substance abuse treatment among persons with comorbid disorders who received mental health care were surprisingly low. As a result, barriers to the adoption of formal recommendations for the treatment of patients with comorbid disorders need to be better understood. ♦

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